

Intégration sur un cycle évanescent

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Source and editorial apparatus

Scope and source

The controlling source is the fifteen-page Number49 scan, printed pages 129–143. The original language is French; the English edition is a standalone translation. The article comprises the introduction, sections 1–3, the *Appendice / Signes*, three bibliography entries and the terminal receipt *Oblatum 30-X-1983*. P. Deligne is the sole printed author. Bibliographic references to other authors are not authorship credits.

The separate collected-paper scan is comparison-only; it never overrides the controlling scan. All supplied audits and prior-work materials remain preserved as inherited evidence. No prior-work branch is accepted on its own authority. The exact archive, source and output identities are recorded in the machine-readable gate inventories accompanying these editions.

Page and copy-matter policy

Each language edition has one output page for each of the fifteen source pages, in source order. The source folio is supplied in the output footer as a locator. Running titles, running bylines, journal masthead and publisher copyright are excluded from the article body. No cover page, missing page, additional author, continuation or terminal publisher matter is invented. The title, sole byline, institutional address, section labels, formulas, appendix, bibliography and receipt are article matter and remain in the readers. Paragraphs continue across source-page boundaries when the source does. Blank remainder after the receipt marks the actual end of the article; it is not a missing continuation.

The first source page prints both “Invent. math. 76, 129–143 (1983)” and “© Springer-Verlag 1984”, with the masthead *Inventiones mathematicae*. Both year statements are preserved here; neither is silently reconciled. The address reads “Institut des Hautes Etudes Scientifiques 35, route de Chartres, F-91440 Bures-sur-Yvette, France”. The unaccented *Etudes* and absence of a comma after *Scientifiques* are retained.

Native mathematics and diagrams

The editions contain editable native mathematics throughout. All eight supplied scanned diagram crops have been replaced by source-addressed native TikZ-CD reconstructions: physical pages 4, 5, 6, 9, 10, 11 (two diagrams) and 14. Objects, labels, arrow directions, equality markers and inclusion relations are checked against direct source renders. No source-image fallback is used. Formula labels remain explicit, including (1), (1’), (1.1.1)–(3.7.2) where printed, and (S.1)–(S.7); ordinary unnumbered displays are not assigned new equation numbers. Display labels are placed on the left, as in the source.

Source-local readings

The following notes record potentially surprising source readings and the sign-sensitive or boundary-sensitive passages. They are not mathematical emendations or proof certification.

Physical 1–3 / printed 129–131

The theorem and corollary have unnumbered printed labels. The contour integral in (1), the orientation of the vanishing cycle, the interval $|t|/R < r < R$ and the two cancelling choices of sign are retained. The source uses $\omega_{X/S} = \pi^! \mathcal{O}_S[1]$ and the basis $dx \wedge dy/dt$. The local trace and the element v_n use cohomological degree 1. The explicit *Notations* paragraph precedes section 1.

Physical 4–5 / printed 132–133

Formula (1.1.1) prints the binomial coefficient $\binom{2n-i}{i}$; this reading is not replaced by an inferred correction. Formula (1.2.1) has the congruence $\mathrm{Tr}(f dx \wedge dy/dt \cdot v_n) \equiv -I(f) \pmod{t^n}$. Formula (1.2.2) visibly prints $I'(\sigma(\lambda, \mu)^* f) = \sigma(\lambda, \mu)^* f$; the right side is not silently changed to an expression involving $I'(f)$. The native local-trace chain retains the lower-case label *tr*. Section 2 begins on physical page 5; Lemma 2.2 continues onto physical page 6.

Physical 6–8 / printed 134–136

The generalized-Jacobian discussion retains $D \neq \emptyset$ and the complete Serre-duality square, with its equality marker, upward right arrow and (*Serre*) label. The continuation uses $H^0(U - D, \omega) = H^0(U - D, \Omega^1)$. The single-prime objects in (2.4.2)–(2.4.3), the sheaf inverse-image symbol $\pi^\bullet$, the additive and multiplicative sequences, $m \leq n$, μ_m and the Lie construction are retained. The source’s comma/semicolon variants in Picard notation are not globally regularized.

Physical 9–11 / printed 137–139

The trace-compatibility diagram has three upper objects, degree 1, two downward arrows and a lower equality. Lemma 2.6 uses its printed three-object H^1 chain. Its proof says “la description de d ”; d is not changed to dw_n . The native base-change triangle on physical page 10 has two converging arrows. On physical page 11 the compactification square retains the distinct prime labels; the induction diagram retains $\supset$ and $\subset$, and the three vertical I arrows. The assumptions $N \geq 3$, the diagonal expression for $I(g)$ and relation (3.6.1) are retained.

Physical 12 / printed 140

The degree bounds use σ , the exponent $g + z$ and weak inequalities. Formula (1') retains its product dot and explicit label. The cycle display prints the final condition $|x_m| = r_n$; it is not silently changed to x_n . The estimate uses $O(p^\beta)$. French “l’une des composante” is retained as printed; normal English grammar does not supply a mathematical emendation.

Physical 13–14 / printed 141–142

Physical page 13 closes section 3.10 and begins the appendix with a separate italic *Signes* heading. The complex notation uses $\text{Hom}^\bullet$. The external products in (S.2) and (S.3) are $\boxtimes_i$, not $\otimes_i$. Formula (S.2) has no terminal full stop; (S.3) does. The iterated-morphism sentence refers to (S.3). Physical page 14 retains the degree shifts, $s^! \rightarrow f_*$, the trace square and its terminal semicolon, the distinct $\mathbf{H}^0$, $H^0\Gamma$ and $H^1\Gamma$ terms, and the distinction between L and L' .

Physical 15 / printed 143

The terminal compatibility statements retain $\text{Tr} \circ \partial = -\text{Res}_x$, the product sign $(-1)^{n_1 n_2}$, the shifted target $\mathbf{H}^0(X(\mathbb{C}), \mathbb{C}[2n])$, the label ε , the factor $(2\pi i)^{-n}$, $U_{0\dots n}$ and $(-1)^{n(n+1)/2}$. The three bibliography entries keep their order and numeric data, followed by the centered roman receipt “Oblatum 30-X-1983” and blank remainder to the actual end of the source.

Independent normalization notes

All modifications made after receipt of the complete supplied state are recorded below. Native diagram reconstruction and layout normalization do not authorize silent changes to printed mathematics.

D042-N001 (physical 1). The generic multi-index is printed as bold Latin n , not Greek ν ; the vector x is bold. The native expression restores these distinctions in both languages.

D042-N002 (physical 1). The diagonal display has no index attached to the summation sign. The supplied subscript n is removed and the source’s bold multi-index **1** is restored in the coefficient.

D042-N003 (physical 3). The defining relation in the Notations paragraph retains the source’s colon-equals sign $X_T := X \times_S T$.

D042-N004 (physical 9). The left-facing arrow in the three-object chain of Lemma 2.6 carries a printed isomorphism marker $\sim$. It is restored; the separate trace-compatibility diagram is not given an additional label.

D042-N005 (physical 6). The Serre-duality prose prints $u : T \rightarrow C$, followed by a section of C_T/T . The unprinted subscript T in the first codomain is removed, preserving the source-local discrepancy.

D042-N006 (physical 7). The construction of $\mathcal{L}_\lambda$ prints the indexed trivial sheaf $\mathcal{L}'_\lambda = \mathcal{O}$. The omitted λ subscript is restored at this occurrence.

D042-N007 (physical 7). The English construction restores the explicit forward reference “see below”, translating the source’s “cf. ci-dessous”.

D042-N008 (physical 8). In paragraph (b), the linear map is printed with the Greek capital Γ , not Latin F . Both languages restore $\Gamma : V \rightarrow W$.

D042-N009 (physical 14). The source distinguishes bold upright $\mathbf{H}$ from ordinary H . Bold $\mathbf{H}$ is restored at exactly three degree-zero supported-hypercohomology occurrences: (S.7) and the first two terms of the following comparison chain. The third term $H^0\Gamma_{\{x\}}$ and later H^1 terms remain ordinary.

D042-N010 (physical 15). The shifted integration morphism and both terms of the displayed ε map use bold upright $\mathbf{H}^0$ in the source; these three instances are restored. The following GAGA equality prints two ordinary italic H^0 instances. That source-local inconsistency is deliberately retained.

Validation scope

The independent gate records source-page replay, cross-language mathematical comparison, native-diagram checks, complete rendered-page layout review and a clean deterministic rebuild followed by nonpatching verification of the frozen output bytes. It does not certify every source glyph, prove the mathematical arguments or verify external historical and bibliographic claims. Source-local anomalies are preserved with notes. All inherited evidence remains available without being promoted to an independent acceptance claim.